Clarity Engine - One-Page App Summary

Repo-based snapshot as of 2026-02-18

What it is

Clarity Engine is a pre-match football intelligence app that predicts match dynamics and explains why.
It combines deterministic data context with narrative analysis and post-match validation loops.

Who it is for

Primary persona: serious bettors who want pre-match edge and evidence-backed narratives.

What it does

- Builds team and player state snapshots per round from FotMob data in PostgreSQL.
- Exposes 14 agent tools for form, style, key players, injuries, H2H, matchups, and scenarios.
- Enforces round-based temporal filtering to prevent data leakage in historical analysis.
- Generates match context and predictions, then evaluates outcomes vs reality after matches.
- Serves API endpoints for fixtures, context, analysis, evaluation, failures, and health checks.
- Provides two operator UIs: Streamlit ops dashboard and a React/Vite web frontend.
- Marks ‘search_news‘ and ‘get_odds‘ as placeholders pending external integration.

How it works (compact architecture from repo evidence)

- Data ingestion: ‘src/data/providers/fotmob.py‘ + ‘scripts/backfill_fotmob.py‘ -> FotMob raw tables.
- State build: ‘scripts/populate_kg_states.py‘ + ‘scripts/populate_player_states.py‘ -> KG state tables.
- Reasoning tools: ‘src/tools/*‘ query PostgreSQL and return structured ‘ToolResponse‘ objects.
- Analysis layer: ‘ContextBuilderV2‘, ‘ClarityEngine‘, and ‘AnalysisEvaluator‘ produce and score outputs.
- Delivery: FastAPI (‘src/api/main.py‘) powers REST endpoints used by ‘src/web/src/lib/api.ts‘.

How to run (minimal getting started)

1. ‘cd /Users/joao/Projects/clarity_engine && source venv/bin/activate‘
2. Install dependencies: ‘pip install -r requirements.txt‘
3. Ensure Postgres is running and ‘DATABASE_URL‘ is set (or local fallback in db config).
4. Quick tool check: ‘python -c "from src.tools import get_team_state; ..."‘
5. Run API: ‘uvicorn src.api.main:app --reload --port 8000‘

Not found in repo

- Single command bootstrap for DB schema + seed data.
- Production deployment/runbook instructions.